RATING METHODOLOGY

12 September 2025

TABLE OF CONTENTS

Scope 1
Rating approach 2
Scorecard 3
Discussion of the scorecard factors 5
Other considerations 9
Scorecard-indicated outcome 12
Assigning issuer-level and instrument-level ratings 13
Key rating assumptions 14
Limitations 14
Moody's related publications 15

Contacts

Luisa De Gaetano +1.212.553.3847
Associate Managing Director
marialuisa.degaetanopolverosi@moodys.com

Caroline Pichon +33.1.5330.1027
Associate Managing Director
caroline.pichon@moodys.com

Jacintha Poh +65.6398.8320
Associate Managing Director
jacintha.poh@moodys.com

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454

Rating Methodology

Gaming

This rating methodology replaces the *Gaming* methodology published on 22 June 2021. We have made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies. These updates do not change our methodological approach.

Scope

This rating methodology applies to companies globally that are primarily engaged in the ownership and operation of gaming outlets, including casinos, online gaming and sports betting.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

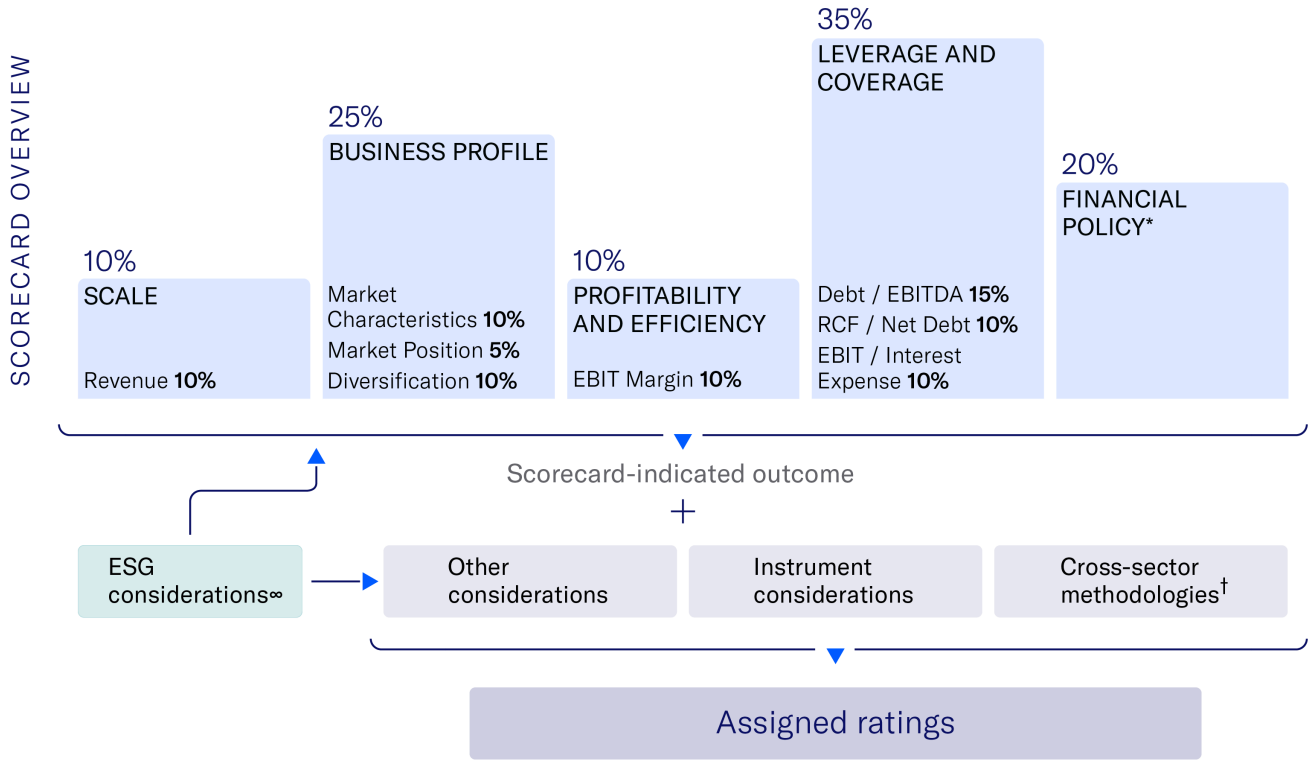
We rate companies that are primarily engaged in the manufacture and distribution of gaming products and technology, such as slot machines and lottery terminals, using our methodology for business and consumer services.[1]

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the gaming rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's Related Publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Gaming scorecard

	SCALE (10%)	BUSINESS PROFILE (25%)			PROFITABILITY and EFFICIENCY (10%)	LEVERAGE and COVERAGE (35%)			FINANCIAL POLICY (20%)
	Revenue (USD Billion)[1] (10%)	Market Characteristics (10%)	Market Position (5%)	Diversification (10%)	EBIT Margin[2] (EBIT / Revenue) (10%)	Debt / EBITDA[3] (15%)	RCF / Net Debt[4] (10%)	EBIT / Interest Expense[5] (10%)	Financial Policy (20%)
Aaa	≥ $50	Gaming demand exceeds supply by a very wide margin and is expected to continue to grow; all gaming facilities are located in close proximity to extremely high-density population areas; regulatory framework is fully developed, extremely predictable and stable, and very unlikely to change, including through legislation.	Company captures all of the market share in all of the markets in which it operates.	Very wide geographic diversification of gaming facilities globally; cash flow contribution is well-balanced across properties, revenue sources and delivery channels (e.g., physical facilities, online gaming, sports betting).	≥ 60%	≤ 0.5x	≥ 75%	≥ 25x	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.
Aa	$20 - $50	Gaming demand exceeds supply by a very wide margin; most gaming facilities are located in close proximity to very high-density population areas; regulatory framework is fully developed, very predictable and stable, and unlikely to change, including through legislation.	Company captures almost all of the market share in the key markets in which it operates.	Wide geographic diversification of gaming facilities globally; cash flow contribution is well-balanced across properties, revenue sources and delivery channels (e.g., physical facilities, online gaming, sports betting).	45% - 60%	0.5x - 1x	55% - 75%	15x - 25x	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.
A	$10 - $20	Gaming demand exceeds supply; most gaming facilities are located in close proximity to high-density population areas; regulatory framework is fully developed, predictable and stable, and unlikely to change, including through legislation.	Company captures a very large market share in the key markets in which it operates.	Wide geographic diversification of gaming facilities across countries within a region; cash flow contribution is well-balanced across properties, revenue sources and delivery channels (e.g., physical facilities online gaming, sports betting).	35% - 45%	1x - 2x	40% - 55%	8x - 15x	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.
Baa	$5 - $10	Gaming demand is equal to supply; gaming facilities are primarily located in close proximity to high-density population areas; regulatory framework is fully developed, somewhat predictable and stable, and has a low probability of change, including through legislation.	Company captures a large market share in the key markets in which it operates.	Wide geographic diversification of gaming facilities across jurisdictions within a country; cash flow contribution is moderately balanced across properties, revenue sources and delivery channels (e.g., physical facilities online gaming, sports betting).	25% - 35%	2x - 3x	25% - 40%	5x - 8x	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.

	SCALE (10%)	BUSINESS PROFILE (25%)			PROFITABILITY and EFFICIENCY (10%)	LEVERAGE and COVERAGE (35%)			FINANCIAL POLICY (20%)
	Revenue (USD Billion)[1] (10%)	Market Characteristics (10%)	Market Position (5%)	Diversification (10%)	EBIT Margin[2] (EBIT / Revenue) (10%)	Debt / EBITDA[3] (15%)	RCF / Net Debt[4] (10%)	EBIT / Interest Expense[5] (10%)	Financial Policy (20%)
Ba	$1.5 - $5	Gaming demand is equal to or modestly less than supply; some gaming facilities are located in close proximity to high-density population areas; regulatory framework is well-developed, although less predictable and stable, or the possibility of unfavorable legislation exists.	Company captures a moderate market share in the key markets in which it operates.	Moderately wide geographic diversification of gaming facilities across jurisdictions within a country; cash flow contribution is somewhat balanced across properties, revenue sources and delivery channels (e.g., physical facilities, online gaming, sports betting).	15% - 25%	3x - 4.5x	15% - 25%	2x - 5x	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.
B	$0.5 - $1.5	Gaming demand is modestly less than supply; gaming facilities are primarily located in close proximity to moderate-density population areas; regulatory framework is developed, but less predictable and stable, or the regulatory environment is consistently challenging and politically charged; likelihood of unfavorable legislation exists.	Company captures a modest market share in the key markets in which it operates.	Narrow geographic diversification of gaming facilities across jurisdictions within a country; cash flow contribution is somewhat unevenly distributed across properties, revenue sources and delivery channels (e.g., physical facilities, online gaming, sports betting).	10% - 15%	4.5x - 6x	5% - 15%	1x - 2x	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.
Caa	$0.25 - $0.5	Gaming demand is less than supply; gaming facilities are primarily located in close proximity to low-density population areas; regulatory framework is not developed, is unclear, or is undergoing substantial change, or regulatory body appears unsupportive, uncertain, or highly unpredictable; likelihood of unfavorable legislation is high.	Company captures a limited market share in the market(s) in which it operates.	Very narrow geographic diversification of gaming facilities across jurisdictions within a country; cash flow contribution comes largely from one or two properties, revenue sources and delivery channels (e.g., physical facilities, online gaming, sports betting).	5% - 10%	6x - 8x	2% - 5%	0.5x - 1x	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.
Ca	< $0.25	Gaming demand significantly less than supply; all gaming facilities are located in close proximity to low density population areas; regulatory framework or regulatory body carry extremely high risk for business continuity.	Company captures very little market share in the market(s) in which it operates.	No geographic diversification of gaming facilities across jurisdictions within a country; cash flow contribution comes largely from one property, revenue source or delivery channel (e.g., physical facilities, online gaming, sports betting).	< 5%	> 8x	< 2%	< 0.5x	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 75%. A value of 75% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is zero. A value of zero equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is 0%. A value of 0% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 35x. A value of 35x or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (10% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

Large-scale gaming companies tend to have greater market share and better access to capital compared with smaller-scale companies. Large companies may also benefit from economies of scale with respect to research and development expenses and corporate overhead. Companies with greater scale generally have lower earnings volatility relative to smaller companies because of the lower risk that a single customer can "take the house" for a large sum with a few significant bets.

A larger scope of operations can reduce a company's reliance on a particular jurisdiction or market. In markets with high barriers to entry, scale may provide a competitive advantage. However, in many regional and local gaming markets, the competitive advantage gained by scale may not be as important because of already low competition.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (25% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Market Characteristics

The characteristics of the markets in which a gaming company operates, and the markets from which it draws customers, are important because they reflect the level of demand for gaming products relative to the supply of customers in the overall market. The competitive and regulatory environments of a market typically affect a company's ability to grow and maintain market share, expand its customer base, generate revenue and remain profitable. A gaming company that operates in or draws customers from one or more

markets located near a large, densely populated area has a larger base of potential customers and typically has a long and consistent history of strong demand for its product offerings. A gaming company that operates in or draws customers from a sparsely populated market is likely to have a smaller customer base, with gaming facilities that are not conveniently located or easily accessible.

A gaming company's ability to capture market share is also dependent on the supply of competitor gaming facilities within the markets in which it operates. A gaming company whose demand exceeds the market supply is in a better position to grow and maintain market share compared with one where supply exceeds demand.

The regulatory framework greatly influences the operating environment of a gaming market. A stable, well-developed regulatory framework that is unlikely to change generally provides a gaming company with greater predictability, allowing it to develop long-term planning to maintain its competitive strength. A regulatory framework that is not fully developed or is unclear, unpredictable or is undergoing substantial change, or a regulatory body that appears unsupportive poses a greater risk to a gaming company's business continuity because of the uncertain operating environment such conditions create. Unfavorable legislative initiatives in the jurisdictions in which a gaming company operates could result in significant negative consequences, such as higher taxes or other restrictions that reduce operating margins or the authorization of additional gaming venues that create increased competition. A regulatory framework may also influence competitive dynamics, i.e., the supply of gaming facilities within the markets in which a gaming company operates.

Market Position

A gaming company's market position is a meaningful indicator of its resilience to economic downturns and intensifying competition. A leading market position may indicate that a company's investments in its facilities and technology are translating into competitive advantages and may provide insight into the extent of its different product offerings. Gaming companies that have stronger market positions tend to be more profitable and generally have more resources to maintain or expand their customer base compared with companies that have weaker market positions. Profitability can vary from one market to another, but it tends to be higher over time in key markets where a company has a leading market position.

Diversification

A gaming company's geographic and product diversification are important because they can mitigate adverse economic trends or changes in consumer habits that affect specific regions or types of delivery channels (e.g., physical facilities or online gaming). Geographic diversification can also mitigate the adverse effects of regulatory changes within a particular market because a company with greater diversity may be able to offset any lost business by relying on markets outside of the affected jurisdiction. A company that offers a broad array of products may appeal to a larger customer base, and a diversified company is more likely to generate stable cash flow across its properties, revenue sources and delivery channels than one that has a narrow geographic or product focus.

How we assess it for the scorecard

We score this factor based on three qualitative subfactors: Market Characteristics; Market Position; and Diversification.

We generally do not expect a given company to match each attribute listed for a given scoring category for each subfactor. We typically assign a subfactor score based on the alpha category for which a company has the greatest number of characteristics. However, there may be cases in which one characteristic is sufficiently important to a particular company's credit profile that it has a large influence on a subfactor score.

Market Characteristics

We score this subfactor based on demand relative to supply on a current and forward-looking basis, the proximity of gaming facilities to densely populated areas and the stability and predictability of the regulatory framework.

In assessing demand, we consider a gaming company's revenue growth. Stable or upward revenue growth typically indicates that demand for a gaming company's product offerings is meeting or exceeding the supply of customers within its markets. Declines in revenue for one or more gaming companies within a market is typically an indicator that the market is saturated, and that there may be weak demand relative to supply.

Our assessment of gaming demand is based on a qualitative assessment of monthly, annual and multi-year trends in gaming revenue. Where available, we rely on information reported by gaming authorities and gaming companies.

We consider the likelihood of ongoing demand for a gaming company's product offerings through a qualitative assessment of population statistics and demographic data in or near the markets in which it operates and from which it attracts customers.

In assessing the regulatory environment in which a gaming company operates, we consider the track record of the regulatory bodies responsible for gaming in each of a gaming company's markets. We assess how fully developed the regulatory framework is, its stability and predictability, and the likelihood that legislative or other changes could alter the competitive landscape in a given market. We also typically assess whether a regulatory regime favors existing market participants or new entrants. Political and cultural changes may also affect the stability and predictability of the regulatory environment, and changes in the regulatory frameworks of neighboring jurisdictions may influence a gaming company's competitive environment in its own market.

Market Position

In assessing market position, we consider how much of the market share a company captures in the markets in which it operates. We typically consider the amount of revenue it collects in a market relative to the market's total gaming revenue, where such data is reported by third-party government or industry entities. We may also consider win per unit, which is an industry metric that shows the average amount of income generated by each slot machine, casino table or gaming device. In markets where aggregated market share data is not reported, we typically use audited financial reports of market participants to estimate the market's total gaming revenue and arrive at a gaming company's market position based on its revenue relative to the total estimated market revenue. For a company that operates in more than one market, our assessment typically centers on the key markets that represent a significant share of a company's revenue.

Diversification

We score this subfactor based primarily on the breadth of a gaming company's geographic presence as well as its cash flow contributions from different revenue sources and delivery channels (e.g., physical facilities or online gaming).

We assess a gaming company's geographic diversification by the number of its gaming facilities and whether they are dispersed across multiple global regions, across countries within a broad geographic region, or across state or local jurisdictions within a country. Companies with operations that are diversified globally typically receive higher scores for this subfactor than companies whose facilities are largely concentrated in one region, country, or state or local jurisdiction.

In assessing diversification across revenue sources and delivery channels, we consider a gaming company's range of distinct product offerings and their proportionate contribution to the company's revenue base and cash flow. A gaming company's product offerings may include traditional on-site casino gaming, online gaming, sports betting or other gaming delivery channels. A company may also offer non-gaming amenities, such as hotels, restaurants, retail stores, convention centers and other types of entertainment, that appeal to a wider array of customers and supplement gaming revenue.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBIT Margin.[3]

EBIT Margin

We use the ratio of earnings before interest and taxes (EBIT) to revenue.

Factor: Leverage and Coverage (35% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: Debt/EBITDA; RCF/Net Debt; and EBIT/Interest Expense.[4]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand business lines, consolidate market positions, advance cost synergies or seek access to new technology.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Social risks to gaming may emerge as consumer preferences evolve and demographics change, which may reduce demand for traditional gaming venues, such as casinos. Consumers, under pressure from weak growth in disposable personal income and increasing living expenses, may limit spending on gaming, a highly discretionary form of entertainment.

Social attitudes toward gambling addiction may increase pressure on gaming companies to adopt more socially responsible measures. Gaming operators may also face reputational risk if they appear to minimize problem gambling or if they appear to market products to vulnerable customers.

Government anti-money laundering efforts may add to gaming companies' compliance costs, thereby pressuring margins. Governments may increase taxes on gaming companies to compensate for reduced tax income elsewhere, resulting in lower profitability.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's business profile. For instance, regulatory authorities may issue additional gaming licenses within a jurisdiction that increase competitive forces for a company. However, we may

assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Cyclical sectors

In a cyclical sector, the scorecard-indicated outcome may be higher than the assigned rating at the top of the economic cycle and lower than the assigned rating at the bottom of the cycle. While using annual or other 12-month-period financials in the scorecard typically provides useful insights into recent or near-term financial results, ratings may also reflect our expectations for the progression of financial results over a longer period that may include a full economic cycle. However, cyclicality itself poses many different types of risks to companies, and cycles do not reverse themselves with predictable regularity. A secular decline or expansion may also affect a cyclical sector. Ratings may incorporate these considerations qualitatively.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

© 2025 Moody's Corporation, Moody's Investors Service, Inc., Moody's Analytics, Inc. and/or their licensors and affiliates (collectively, "MOODY'S"). All rights reserved.
CREDIT RATINGS ISSUED BY MOODY'S CREDIT RATINGS AFFILIATES ARE THEIR CURRENT OPINIONS OF THE RELATIVE FUTURE CREDIT RISK OF ENTITIES, CREDIT COMMITMENTS, OR DEBT OR DEBT-LIKE SECURITIES, AND MATERIALS, PRODUCTS, SERVICES AND INFORMATION PUBLISHED OR OTHERWISE MADE AVAILABLE BY MOODY'S (COLLECTIVELY, "MATERIALS") MAY INCLUDE SUCH CURRENT OPINIONS. MOODY'S DEFINES CREDIT RISK AS THE RISK THAT AN ENTITY MAY NOT MEET ITS CONTRACTUAL FINANCIAL OBLIGATIONS AS THEY COME DUE AND ANY ESTIMATED FINANCIAL LOSS IN THE EVENT OF DEFAULT OR IMPAIRMENT. SEE APPLICABLE MOODY'S RATING SYMBOLS AND DEFINITIONS PUBLICATION FOR INFORMATION ON THE TYPES OF CONTRACTUAL FINANCIAL OBLIGATIONS ADDRESSED BY MOODY'S CREDIT RATINGS. CREDIT RATINGS DO NOT ADDRESS ANY OTHER RISK, INCLUDING BUT NOT LIMITED TO: LIQUIDITY RISK, MARKET VALUE RISK, OR PRICE VOLATILITY. CREDIT RATINGS, NON-CREDIT ASSESSMENTS ("ASSESSMENTS"), AND OTHER OPINIONS INCLUDED IN MOODY'S MATERIALS ARE NOT STATEMENTS OF CURRENT OR HISTORICAL FACT. MOODY'S MATERIALS MAY ALSO INCLUDE QUANTITATIVE MODEL-BASED ESTIMATES OF CREDIT RISK AND RELATED OPINIONS OR COMMENTARY PUBLISHED BY MOODY'S ANALYTICS, INC. AND/OR ITS AFFILIATES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT CONSTITUTE OR PROVIDE INVESTMENT OR FINANCIAL ADVICE, AND MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT AND DO NOT PROVIDE RECOMMENDATIONS TO PURCHASE, SELL, OR HOLD PARTICULAR SECURITIES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT COMMENT ON THE SUITABILITY OF AN INVESTMENT FOR ANY PARTICULAR INVESTOR. MOODY'S ISSUES ITS CREDIT RATINGS, ASSESSMENTS AND OTHER OPINIONS AND PUBLISHES OR OTHERWISE MAKES AVAILABLE ITS MATERIALS WITH THE EXPECTATION AND UNDERSTANDING THAT EACH INVESTOR WILL, WITH DUE CARE, MAKE ITS OWN STUDY AND EVALUATION OF EACH SECURITY THAT IS UNDER CONSIDERATION FOR PURCHASE, HOLDING, OR SALE.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS, AND MATERIALS ARE NOT INTENDED FOR USE BY RETAIL INVESTORS AND IT WOULD BE RECKLESS AND INAPPROPRIATE FOR RETAIL INVESTORS TO USE MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS OR MATERIALS WHEN MAKING AN INVESTMENT DECISION. IF IN DOUBT YOU SHOULD CONTACT YOUR FINANCIAL OR OTHER PROFESSIONAL ADVISER.
ALL INFORMATION CONTAINED HEREIN IS PROTECTED BY LAW, INCLUDING BUT NOT LIMITED TO, COPYRIGHT LAW, AND NONE OF SUCH INFORMATION MAY BE COPIED OR OTHERWISE REPRODUCED, REPACKAGED, FURTHER TRANSMITTED, TRANSFERRED, DISSEMINATED, REDISTRIBUTED OR RESOLD, OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE, IN WHOLE OR IN PART, IN ANY FORM OR MANNER OR BY ANY MEANS WHATSOEVER, BY ANY PERSON WITHOUT MOODY'S PRIOR WRITTEN CONSENT. FOR CLARITY, NO INFORMATION CONTAINED HEREIN MAY BE USED TO DEVELOP, IMPROVE, TRAIN OR RETRAIN ANY SOFTWARE PROGRAM OR DATABASE, INCLUDING, BUT NOT LIMITED TO, FOR ANY ARTIFICIAL INTELLIGENCE, MACHINE LEARNING OR NATURAL LANGUAGE PROCESSING SOFTWARE, ALGORITHM, METHODOLOGY AND/OR MODEL.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT INTENDED FOR USE BY ANY PERSON AS A BENCHMARK AS THAT TERM IS DEFINED FOR REGULATORY PURPOSES AND MUST NOT BE USED IN ANY WAY THAT COULD RESULT IN THEM BEING CONSIDERED A BENCHMARK.
All information contained herein is obtained by MOODY'S from sources believed by it to be accurate and reliable. Because of the possibility of human or mechanical error as well as other factors, however, all information contained herein is provided "AS IS" without warranty of any kind. MOODY'S adopts all necessary measures so that the information it uses in assigning a credit rating is of sufficient quality and from sources MOODY'S considers to be reliable including, when appropriate, independent third-party sources. However, MOODY'S is not an auditor and cannot in every instance independently verify or validate information received in the credit rating process or in preparing its Materials.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability to any person or entity for any indirect, special, consequential, or incidental losses or damages whatsoever arising from or in connection with the information contained herein or the use of or inability to use any such information, even if MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers is advised in advance of the possibility of such losses or damages, including but not limited to: (a) any loss of present or prospective profits or (b) any loss or damage arising where the relevant financial instrument is not the subject of a particular credit rating assigned by MOODY'S.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability for any direct or compensatory losses or damages caused to any person or entity, including but not limited to by any negligence (but excluding fraud, willful misconduct or any other type of liability that, for the avoidance of doubt, by law cannot be excluded) on the part of, or any contingency within or beyond the control of, MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers, arising from or in connection with the information contained herein or the use of or inability to use any such information.
NO WARRANTY, EXPRESS OR IMPLIED, AS TO THE ACCURACY, TIMELINESS, COMPLETENESS, MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE OF ANY CREDIT RATING, ASSESSMENT, OTHER OPINION OR INFORMATION IS GIVEN OR MADE BY MOODY'S IN ANY FORM OR MANNER WHATSOEVER.
Moody's Investors Service, Inc., a wholly-owned credit rating agency subsidiary of Moody's Corporation ("MCO"), hereby discloses that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by Moody's Investors Service, Inc. have, prior to assignment of any credit rating, agreed to pay Moody's Investors Service, Inc. for credit ratings opinions and services rendered by it. MCO and all MCO entities that issue ratings under the "Moody's Ratings" brand name ("Moody's Ratings"), also maintain policies and procedures to address the independence of Moody's Ratings' credit ratings and credit rating processes. Information regarding certain affiliations that may exist between directors of MCO and rated entities, and between entities who hold credit ratings from Moody's Investors Service, Inc. and have also publicly reported to the SEC an ownership interest in MCO of more than 5%, is posted annually at ir.moodys.com under the heading "Investor Relations — Corporate Governance — Charter and Governance Documents - Director and Shareholder Affiliation Policy."
Moody's SF Japan K.K., Moody's Local AR Agente de Calificación de Riesgo S.A., Moody's Local BR Agência de Classificação de Risco LTDA, Moody's Local MX S.A. de C.V, I.C.V., Moody's Local PE Clasificadora de Riesgo S.A., and Moody's Local PA Calificadora de Riesgo S.A. (collectively, the "Moody's Non-NRSRO CRAs") are all indirectly wholly-owned credit rating agency subsidiaries of MCO. None of the Moody's Non-NRSRO CRAs is a Nationally Recognized Statistical Rating Organization.
Additional terms for Australia only: Any publication into Australia of this document is pursuant to the Australian Financial Services License of MOODY'S affiliate, Moody's Investors Service Pty Limited ABN 61 003 399 657AFSL 336969 and/or Moody's Analytics Australia Pty Ltd ABN 94 105 136 972 AFSL 383569 (as applicable). This document is intended to be provided only to "wholesale clients" within the meaning of section 761G of the Corporations Act 2001. By continuing to access this document from within Australia, you represent to MOODY'S that you are, or are accessing the document as a representative of, a "wholesale client" and that neither you nor the entity you represent will directly or indirectly disseminate this document or its contents to "retail clients" within the meaning of section 761G of the Corporations Act 2001. MOODY'S credit rating is an opinion as to the creditworthiness of a debt obligation of the issuer, not on the equity securities of the issuer or any form of security that is available to retail investors.
Additional terms for India only: Moody's credit ratings, Assessments, other opinions and Materials are not intended to be and shall not be relied upon or used by any users located in India in relation to securities listed or proposed to be listed on Indian stock exchanges.
Additional terms with respect to Second Party Opinions and Net Zero Assessments (as defined in Moody's Ratings Rating Symbols and Definitions): Please note that neither a Second Party Opinion ("SPO") nor a Net Zero Assessment ("NZA") is a "credit rating". The issuance of SPOs and NZAs is not a regulated activity in many jurisdictions, including Singapore. JAPAN: In Japan, development and provision of SPOs and NZAs fall under the category of "Ancillary Businesses", not "Credit Rating Business", and are not subject to the regulations applicable to "Credit Rating Business" under the Financial Instruments and Exchange Act of Japan and its relevant regulation. PRC: Any SPO: (1) does not constitute a PRC Green Bond Assessment as defined under any relevant PRC laws or regulations; (2) cannot be included in any registration statement, offering circular, prospectus or any other documents submitted to the PRC regulatory authorities or otherwise used to satisfy any PRC regulatory disclosure requirement; and (3) cannot be used within the PRC for any regulatory purpose or for any other purpose which is not permitted under relevant PRC laws or regulations. For the purposes of this disclaimer, "PRC" refers to the mainland of the People's Republic of China, excluding Hong Kong, Macau and Taiwan.

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the issuer/deal page on https://ratings.moodys.com for the most updated credit rating action information and rating history.

REPORT NUMBER 1450288

Author

Geordie Thompson

CLIENT SERVICES

Americas	1-212-553-1653
Asia Pacific	852-3551-3077
Japan	81-3-5408-4100
EMEA	44-20-7772-5454